IMPACT OF GENERATIVE AI ON PROMOTING CREATIVITY AND INNOVATION ON SECONDARY LEVEL STUDENT

A Research Proposal

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CHAPTER-I

Introduction

1.1 Background of the study

Creativity and innovation are now seen as core to personal development, 21st-century education, and future careers. Traditional secondary education, however, has often constrained creative thinking: teacher-directed classrooms, lecture-based delivery, and standardized assessment tend to prioritize rote memorization, compliance, and factual recall over open-ended exploration, problem-finding, and divergent thinking. Under these conditions, students often have less room to develop creative agency and confidence. In response, modern educational frameworks and national policies increasingly call for active, student-centered learning environments that build creativity, problem-solving, and digital literacy directly into instruction. Generative Artificial Intelligence (GenAI) has advanced quickly and become widely accessible, and it is starting to change how teaching happens in secondary classrooms. GenAI tools are no longer just administrative aids; they act as active partners in learning, supporting deeper cognitive engagement. Through large language models, visual generative engines, and domain-specific adaptive platforms, GenAI can scaffold learners within their Zone of Proximal Development (ZPD) using natural language and iterative feedback. At the secondary level, where students begin encountering more abstract and complex material, GenAI changes the usual dynamic between student, technology, and teacher. Some studies suggest GenAI tools can support the different dimensions of creativity — fluency, flexibility, originality, and elaboration. Acting as co-creators or brainstorming partners, these tools can lower entry barriers such as language anxiety, drawing limitations, or coding frustration, freeing students to focus on conceptual depth, narrative design, and iterative prototyping. Some research reports positive effects on students' divergent thinking, self-efficacy, vocabulary, and originality in final work, though the studies behind these claims are not cited here. At the same time, GenAI use in secondary education carries real risk. There's a tension between AI as scaffold and AI as substitute: when students accept AI outputs without evaluating them, learning can shift into passive "cognitive offloading." Without deliberate pedagogical structure, students risk becoming dependent on automated systems — leading to shallow or simulated understanding, shorter attention spans, and over-reliance on AI-generated suggestions. Heavy AI dependency has also been linked to lower critical

thinking scores and weaker independent problem-solving. Existing research on GenAI's educational impact is also fragmented, both in method and geography. About 80% of school-level AI studies focus on secondary education, but most of this research evaluates GenAI's raw computational capabilities or reports descriptive attitudes rather than how it integrates into actual curricula and teaching. Few studies use classroom observation, longitudinal designs, or standardized rubrics to track how GenAI affects student creativity over time. These gaps point to a clear need: research into how secondary students negotiate agency, critical literacy, and original expression in AI-augmented learning environments.

1.2 Statement of the Problem

Generative Artificial Intelligence (GenAI) has spread quickly through secondary education, but pedagogical models for using it well have not kept pace. GenAI tools are often described as "cognitive amplifiers" that lower barriers to student engagement, but deploying them without scaffolding raises cognitive, instructional, socio-cultural, and methodological problems. The core issue is the Scaffolding vs. Substitution Paradox (Cognitive Offloading & Fixation). When students accept AI outputs without evaluating them, learning can shift from active construction to passive consumption. This creates an "Affective–Cognitive Gap," where students gain confidence and self-efficacy without developing deep conceptual or metacognitive understanding. By offloading effortful tasks and syntax-related struggle, students may skip the "desirable difficulties" and "productive struggle" that authentic creative reasoning depends on. Unguided use can also lead to "cognitive fixation," narrowing independent thinking and pushing students toward similar, less original outputs. There is also a Teacher-Led Mediation Void and TPK Deficit. Successful integration of GenAI depends on teacher readiness and specialized Technological Pedagogical Content Knowledge (TPACK), but a "pedagogical adoption gap" means many teachers lack frameworks for structuring human-AI collaboration. Without teacher-led scaffolding and "human-in-the-loop" (HITL) checkpoints, students may skip structured processes like the "Prompt-Reflect-Verify" loop — justifying, critiquing, and editing AI suggestions. Left unaddressed, GenAI use risks becoming a form of "Automated Domestication," reducing student agency and long-term critical thinking. The problem extends to Socio-Linguistic Bias and Algorithmic Cultural Colonization. Commercial GenAI platforms and large language models are often trained on Western-centric datasets, Eurocentric academic

norms, and dominant linguistic varieties. Used without critical guidance in diverse classrooms, these systems can marginalize indigenous knowledge frameworks and regional linguistic variation, and create difficulties for non-native English speakers and students working in regional syntactic structures. Related to this are "pedagogical hallucinations," where AI-generated lesson frameworks assume access to high-bandwidth infrastructure, smartboards, and commercial resources that under-resourced schools may not have. Research on GenAI also has methodological gaps — what could be called a K-12 Empirical Deficit. Much of the literature carries an "adult-learning bias," with a large share of empirical studies conducted in higher education. Findings from self-regulated university undergraduates may not generalize to secondary students, who differ in cognitive maturity, attention regulation, and digital literacy. The limited K–12 research that exists is fragmented, often based on short-term pilots, workshops, or single-semester designs, and says little about long-term skill retention, cognitive ergonomics, or a possible "plateau effect" once the novelty of the technology wears off. Much of it also relies on subjective self-report surveys, which can be skewed by social desirability and novelty-driven satisfaction. This study investigates GenAI's impact on creativity and innovation among secondary students, examining cognitive, instructional, socio-cultural, and methodological dimensions. It looks at how GenAI can function as a cognitive scaffold rather than a substitute, the role of teacher-led mediation and TPACK, the value of human-in-the-loop approaches, and how socio-linguistic and contextual factors shape student creativity. The findings are intended for teachers, students, researchers, policymakers, curriculum developers, and other stakeholders working on the responsible integration of GenAI in secondary education.

1.3 Objectives of the Study

The objectives of the study are:

a. To evaluate the impact of GenAI tools on student creativity and cognitive engagement — measuring changes in fluency, flexibility, originality, and elaboration, and comparing how unguided AI access versus structured, human-in-the-loop scaffolding (such as a "Prompt-Reflect-Verify" loop) affects independent problem-solving and the risk of cognitive dependency.

b. . To design and validate a teacher-mediated instructional framework for equitable human-AI collaboration — examining the role of teacher AI literacy (TPACK) and addressing socio-linguistic and cultural bias, to build a practical, curriculum-aligned

model that combines GenAI with active approaches such as Project-Based Learning or Design Thinking.

1.4 Research Questions

The following research questions will guide this study:

a. To what extent do different modalities of Generative AI (specifically text-based large language models and visual generative systems) impact secondary level students' creative thinking performance across the multidimensional constructs of fluency, flexibility, originality, and elaboration?
b. How does the implementation of structured pedagogical scaffolding (such as human-in-the-loop "Prompt-Reflect-Verify" guidelines) compare with unguided, passive AI access in mitigating the risks of cognitive offloading, mental fixation, and automated dependency among secondary students ?
c. How do teacher-led mediation practices and educators' Technological Pedagogical Content Knowledge (TPACK) moderate the instructional quality of GenAI-supported lessons, and how do they help mitigate algorithmic bias and preserve students' linguistic and cultural authenticity?

1.5 Definition of the Key Terms

The key terms are defined as following for the better understanding in a consistent manner through the study:

a. To ensure conceptual clarity and a standardized interpretation of the core constructs investigated in this study, the following key terms are operationally defined: Generative Artificial Intelligence (GenAI): A class of advanced computational agents driven by large language models (LLMs) and deep learning architectures that can dynamically generate fluent, multimodal outputs—including text, images, video, and code—from natural language inputs . In this study, GenAI platforms are conceptualized not as passive, neutral tools, but as active cognitive co-creators and collaborative partners that restructure classroom ideation and problem-solving .
b. Creativity / Creative Thinking: A multidimensional cognitive competency defined as the execution of divergent thinking performance . For the purpose of this research, it is measured across four distinct constructs: fluency (the volume of ideas generated), flexibility (the ability to pivot between different conceptual categories), originality (the

statistical uniqueness and novelty of ideas), and elaboration (the amount of detail, structure, and complexity built into final products) .

c. Cognitive Offloading: The psychological and behavioral phenomenon of delegating effortful mental tasks, memory storage, or reasoning processes to external technological devices . In this study, unguided offloading is linked to cognitive dependency, where students bypass the necessary cognitive struggle and "desirable difficulties" required to build independent critical thinking and strategic internalization .
d. Technological Pedagogical Content Knowledge (TPACK / Intelligent-TPACK): An established theoretical framework representing the dynamic, intersecting knowledge bases (content, pedagogy, and technology) that educators must synthesize to design effective, technology-integrated lessons . This study specifically utilizes Intelligent-TPACK, which extends the traditional model to include specialized competencies in AI literacy, critical prompt design, and the ethical mediation of algorithmic agents .
e. Pedagogical Scaffolding / Human-in-the-Loop (HITL): A systematic instructional methodology that embeds technology-mediated tasks within structured reflective cycles . Operationally, this refers to teacher-facilitated practices (such as the "Prompt-Reflect-Verify" loop) that intentionally re-introduce cognitive friction, requiring students to actively critique, justify, and edit AI-generated outputs rather than copying them passively .
f. Socio-Linguistic and Algorithmic Bias: The systemic, value-laden disparities embedded within GenAI platforms due to the demographic, cultural, and linguistic composition of their underlying training data . In educational settings, this bias manifests as cultural stereotyping, the structural penalization of non-native English syntactical structures, and high false-positive error rates among automated AI detectors when evaluating minoritized or multilingual student cohorts .
g. Secondary Level Students: Learners formally enrolled in secondary compulsory schooling contexts (typically ages 12–18) . This cohort represents a distinct developmental stage of adolescence characterized by rapid psychological, behavioral, and emotional transitions, during which students are actively shaping their academic self-efficacy, independent reasoning, and digital identity profiles.

CHAPTER II

2.1. Review of related literature and theoretical framework

In this chapter elaborates and critically analyze the existing literature on a particular topic or research question. It is an essential component of any research project that provides a comprehensive understanding of the current state of knowledge, skill and practices on the research topics. This section consists of the review of the theoretical, empirical and implication review for the research study and conceptual framework.

2.2 Review of the theoretical Literature

Torrance (1987), The Torrance Test of Creative Thinking (TTCT), identifies and evaluates creativity potential through two parts: a verbal test and a figural test. The verbal test includes Asking Questions, Guessing Causes, Guessing Consequences, Product Improvement, and Unusual Uses. The figural test includes Picture Construction, Picture Completion, and Lines or Circles Completion.

Looney (2009), Assessment and Innovation in Education, looks at how summative assessment affects innovation. Innovation has been an important topic in education, and successful innovation depends on human creativity, knowledge, skills, and talents — largely nurtured and developed through education.

Cerna (2014), The Government of Innovation in Education, defines innovation as the implementation of improved ideas, knowledge, and practices.

Nath and Iswary (2015), The World Wide Web as the Largest Global Information Media, describe the World Wide Web as a global information medium through which users share, read, and write data using computers connected to the Internet. They also discuss Web 3.0 as the emergence of the semantic web, where computers generate and process new information rather than relying entirely on humans.

Rudman and Bruwer (2016) explain that the Web consists of documents connected to the World Wide Web, accessible worldwide, that provide information on particular topics. They add that Web 3.0 involves an integrated web experience in which machines understand and catalogue data in ways similar to humans.

Tang (2017), Handbook of the Management of Creativity and Innovation, notes that the word creativity comes from the Latin creare, meaning "to make," and is also linked to the ideas of arising or growing.

Shubina (2019), The Research Pattern of Creativity and Innovation: The Period of 2010–2019, identifies and synthesizes findings across studies of creativity and innovation, drawing on papers published in highly ranked and cited journals indexed in the Web of Science Core Collection. The number of journals and publications on creativity and innovation grew significantly from 2010 to 2019, especially by 2019.

Vincent-Lancrin (2019), Measuring Innovation in Education 2019, describes innovation as a new or improved product or process, or a combination of both, that differs significantly from an organization's previous products or processes and has been made available to potential users or brought into use. The discussion stresses the need for shared standards on how such products or processes are developed and used.

Psychol (2022), What Do Educators Need to Know About the Torrance Tests of Creative Thinking: A Comprehensive Review, offers school psychologists, educators, and curriculum designers a practical guide to the TTCT, developed by E. Paul Torrance in the 1960s, as a measure of divergent thinking skills.

Brit (2023), Creativity Skills, defines creativity as the ability to think about a task or problem in a new or different way, and to use imagination to generate new ideas.

Thompson (2024), What Is Innovation in Education and Why Is It Important?, explains that innovation means making a change or doing something in a new way. It doesn't require invention — creativity and adaptability are part of the process. One example given: students watch lectures at home and complete assignments in the classroom.

Naiman and Brinpickings (2024), Creativity at Work, describe creativity as the ability to draw on knowledge, insight, information, inspiration, and fragments of ideas already in the mind — a process of bringing something previously hidden into existence and giving it new life.

Kerr (2024), Creativity, defines creativity as the ability to make or bring something new into existence, whether a new solution to a problem, a new method or device, or a new artistic object or form.

2.3. Some of the Generative AI tools used in education

1. Conversational Agents and Large Language Models (LLMs):

   Conversational Agents and Large Language Models (LLMs) These are AI tools that let students and teachers interact with AI through natural language — for brainstorming,

answering questions, explaining concepts, analyzing text, generating ideas, and supporting teaching and learning tasks. They give personalized responses and immediate feedback based on what the user asks. Examples include:

a. ChatGPT

b. Google Gemini/Bard

c. Perplexity AI

d. Claude AI

e. Doubao

f. Other LLM-based educational applications.

2. Multimodal Generative AI Platforms:

   These generate text, images, audio, music, and video from user prompts. They support creativity, digital storytelling, graphic design, presentations, and multimedia projects, and cut down the technical work involved in producing content, so students can focus more on ideas and critical thinking. Examples include:

   a. Midjourney

   b. DALL·E

   c. Leonardo AI

   d. Stable Diffusion

   e. Entry etc

3. Domain-Specific Simulation and Interactive Web Applications:

   These are educational technology tools built to support learning in specific subject areas through interactive activities, simulations, problem-solving, and virtual experiments. They help students grasp complex or abstract concepts by letting them interact with material directly and see the results of their actions. Examples include:

   a. QANDA

   b. DIKSHA

   c. PRIMARYAI

   d. PoseBlocks

   e. Google Teachable Machine etc

4. Block-Based AI and Visual Programming:

   These tools use drag-and-drop blocks instead of standard programming syntax, which makes them useful for introducing children and school students to programming, AI, machine learning, algorithms, and computational thinking. They lower the barrier of

traditional coding so students can focus on logic and problem-solving. Examples include:

a. Scratch

b. PopBots

c. Snap!

d. NetsBlox

e. MIT App Inventor

f. PictoBlox

g. LearningML

h. VotestratesML.

5. Intelligent Writing and Automated Feedback Assistants:

   These AI-powered tools help students and teachers with writing, grammar, vocabulary, paraphrasing, and other language skills. They also give personalized feedback and help teachers prepare lesson plans, worksheets, assessments, and other materials. Examples include:

   a. QuillBot

   b. Grammarly

   c. MagicSchool AI

   d. Conker

   e. Eduaide.AI

   f. Diffit

   g. Brisk Teaching

   h. TeacherMatic

   i. Curipod

   j. Shikshana Foundation

The main purpose of study is to find and investigate the impact of Generative AI tools on promoting creativity and innovation in the classroom. The Generative AI has potential impact on developing creativity and innovative thought among the student. It provides new and unique opportunities to express and explore their ideas and collaborate with others.

2.4Review of the empirical Literature

Abdellatif et al. (2026), in a systematic review following the PRISMA 2020 guidelines, examined the effectiveness of Artificial Intelligence (AI)-supported interventions in promoting student creativity. The review synthesized evidence from 19 high-quality experimental and quasi-experimental studies involving 1,695 participants from elementary to higher education. Structured AI interventions had a strong positive effect on students' creative thinking, with overall Cohen's *d* values frequently exceeding 1.0, though the effect varied across dimensions of creativity. Fluency showed the strongest improvement, with increases ranging from 35% to 99%. Abu Owda et al. (2023), for example, reported that students' fluency scores rose from 14.04 before the intervention to 27.92 after. Flexibility also improved considerably, by 19.6% to 25.6%. Originality and elaboration were more context-dependent: Abu Owda et al. (2023) reported a 108% increase in originality scores after a structured intervention, while Toma and Yánez-Pérez (2024) found that 25% of undergraduate participants showed a decrease in divergent-thinking scores after using ChatGPT without instructional guidance. Together, these findings suggest AI supports student creativity when integrated through structured, guided activities, while unguided use may reduce independent imagination and creative thinking.

Finger et al. (2026) conducted a systematic review of 16 high-quality empirical studies published between 2021 and 2025 on the integration of AI into Project-Based Learning (PjBL), focusing on student-centered and active learning environments. AI-enhanced PjBL produced a large positive effect on students' conceptual understanding (Cohen's $d = 1.30$), and a strong positive relationship emerged between students' digital literacy and creativity ($r = 0.714$). The researchers explained that when activities require students to critically evaluate AI-generated information, students develop both digital literacy and creative thinking together — a cycle of human-AI collaboration in which students question, evaluate, modify, and improve AI-generated ideas. But they also flagged a risk of over-dependence: when students simply copy and paste AI-generated answers without evaluating them, cognitive offloading can occur, weakening independent problem-solving. The study stressed the need for teacher guidance and instructional scaffolding when AI is used in PjBL.

A systematic review of 72 empirical studies published between 2014 and 2024 examined the use and development of Intelligent Learning Systems (ILS) in K-12 education, finding a rapid rise in research on intelligent educational technologies, especially post-pandemic. About 72% of the publications fell between 2020 and 2024, a 215% increase in the annual publication average compared with the pre-pandemic period. Intelligent Tutoring Systems were the most studied type of ILS, at 43.1% (n = 31), followed by Adaptive Learning Systems at 20.8% (n = 15) and Learning Analytics Tools at 15.3% (n = 11). Secondary education accounted for the largest share of implementations, at 52.8% (n = 38), with primary education at 33.3% (n = 24). Mathematics was the most commonly studied subject (45.8%, n = 33), followed by Science (23.6%, n = 17). The review concluded that ILS can offer personalized learning, flexible pacing, and immediate feedback, though effectiveness depends on curriculum alignment, teacher preparation, and maintaining meaningful teacher-student interaction.

Existing empirical research on AI and educational technology has several methodological limitations. One is the dominance of higher education: a comprehensive meta-analysis found that 95.5% (k = 21) of experimental studies on ChatGPT's educational impact were conducted in tertiary higher education, with only one focused on secondary school learners. Higher education similarly accounted for 52.6% of the studies in the creativity review and 56% of the studies in the ILS review. This imbalance makes it hard to generalize from university students to secondary school students, since learners at different levels differ in cognitive development, learning experience, and digital literacy.

A second limitation is a high risk of assessment bias and subjectivity. Many studies don't use standardized, validated assessment instruments — in systematic reviews of digital multimodal composing, more than 80% of the synthesized studies carried a high risk of rater bias because they relied on single-rater subjective assessments without reporting inter-rater reliability. Research on K-12 digital comics found that 22 of 30 empirical evaluations didn't report effect sizes at all, relying instead on descriptive findings and student satisfaction surveys.

A third limitation is the lack of longitudinal research. In the ILS review, 87% of the 72 studies included no longitudinal follow-up, and the median intervention lasted only six weeks. That leaves limited evidence on whether gains in creativity, motivation, and learning outcomes hold up over time — and raises the possibility that some findings simply reflect a novelty effect, where students engage more with a new technology because it's new, not because it works better.

Taken together, this body of research points to real potential for AI and Web 3.0 tools to support creativity, innovation, conceptual understanding, digital literacy, and problem-solving — but with a caveat that runs through nearly every study reviewed here: the evidence comes overwhelmingly from higher education, from short interventions, and from studies using subjective, single-rater assessments. Secondary-school learners in particular are underrepresented, which is the clearest gap for future research to address, alongside longer-term studies that can show whether these creativity gains actually last.

CHAPTER III

3.1 Methods And Procedures

This chapter describes the research methods and procedures used to investigate the impact of Generative Artificial Intelligence (GenAI) on creativity and innovation among secondary-level students. It addresses the gap the literature identifies between technological integration and measurable learner outcomes, moving from description toward an empirical, classroom-based methodology.

The procedures evaluate how different AI modalities — text-based large language models and visual generative systems — affect the multiple dimensions of creative thinking under varying instructional scaffolding conditions. This includes the structure of the classroom-embedded intervention: participant recruitment, tool validation, baseline measurement, pedagogical orchestration, and post-intervention assessment

3.2 Research Design and Methods of Study

This study adopted a **sequential explanatory mixed-methods quasi-experimental design** to capture both the quantitative changes in students' creative performance and the qualitative aspects of human-AI cognitive collaboration.

The quantitative strand used a pre-test/post-test control group design to measure changes in specific dimensions of creativity: fluency, flexibility, originality, and elaboration. The qualitative strand incorporated classroom observations, student interaction logs, and semi-structured interviews to explain the cognitive pathways, mental models, and "Vicious or Virtuous Cycles" that emerged during human-AI co-creation.

A quasi-experimental design was necessary here because randomly assigning individual students would have been disruptive to existing school timetables and classroom cohorts. By keeping intact classroom units and purposively assigning them to contrasting treatment conditions, the study preserved ecological validity while controlling for external classroom variables.

The study ran in three phases: baseline measurement through pre-testing, a six-week intervention using pedagogical scaffolding, and post-testing with interviews. The experimental group followed the "Prompt-Reflect-Verify" framework; the control group had un-scaffolded access to GenAI tools.

3.3 Methodology

The study combined quantitative and qualitative data collection and analysis, using survey and empirical research approaches. Students were divided into two comparable groups and worked through a classroom-embedded intervention.

The project-based learning curriculum itself ran for six weeks, within a broader eight-week data collection cycle. The sample population consisted of secondary-level students and their classroom teachers in public secondary schools. Data came from

standardized assessments, Likert-scale instruments, classroom observations, digital interaction logs, creative multimodal artifacts, and semi-structured interviews.

Quantitative data were analyzed using SPSS; qualitative data using NVivo. Quantitative analysis included paired-sample t-tests, ANCOVA, Hedges' *g*, and Cohen's *d*. Qualitative analysis followed deductive and inductive coding and thematic synthesis, with inter-coder reliability assessed using Cohen's Kappa.

3.3.1 Sample

The target population consisted of secondary-level students, typically 14–16 years old, and their classroom teachers in public secondary schools.

The final sample comprised **N = 240 students** and **N = 8 teachers** across eight intact classrooms, split evenly between conditions: **n = 120** in the experimental group and **n = 120** in the control group, four intact classrooms per condition. Four teachers were assigned to each condition.

The student cohort was Grade 9, aged 14–16. The primary AI tools deployed were ChatGPT-4o, Midjourney, and Canva AI. Teachers used MagicSchool AI and lesson-planning LLMs as their primary AI tools

3.3.2 Sampling strategy

This study used a **two-stage purposive and cluster sampling strategy** to balance methodological feasibility with representative demographic diversity.

In the first stage, schools were selected purposively based on stable digital infrastructure: one-to-one student laptop or tablet access, stable high-speed internet, and teachers with basic digital literacy. This controlled for first-level digital-divide barriers — hardware and connectivity access — that could otherwise compromise implementation fidelity.

In the second stage, intact Grade 9 classrooms within the selected schools were treated as clusters and randomly assigned to the experimental or control condition. Classrooms were paired by baseline academic performance and pre-intervention demographic balance to minimize selection bias.

3.3.3 Research tools

This study used a suite of psychometrically validated, standardized instruments alongside qualitative tracking tools to capture the intervention's multidimensional outcomes:

1. **Torrance Tests of Creative Thinking (TTCT – Figural and Verbal)**
2. **Self-Directed Learning Instrument (SDLI)**
3. **Meta AI Literacy Scale (MAILS) Subscale**
4. **Structured Classroom Observation Protocol**
5. **Digital Prompt-Engineering Logs**
6. **Student Interaction Logs**

7. **Semi-Structured Interview Schedule**
8. **Multimodal Product Evaluation Rubric**

The **TTCT (Figural and Verbal)** was the primary measure of student creativity, covering *fluency* (number of relevant ideas), *flexibility* (range of categories), *originality* (uniqueness), and *elaboration* (amount of detail).

The **SDLI**, grounded in Zimmerman's self-regulation model, used a 5-point Likert scale to assess students' metacognitive monitoring, planning, and learning motivation.

The **MAILS Subscale** evaluated students' digital and AI self-efficacy and their critical capability to evaluate and navigate automated systems, also on a 5-point Likert scale.

A **Structured Classroom Observation Protocol** — a researcher-developed checklist — recorded the frequency and quality of student-AI dialogic turn-taking, independent problem-solving struggles, and teacher-student scaffolding loops, with Cohen's Kappa applied to rater logs for reliability.

3.3.4 Source of data

Data for this study came from multiple actors and artifacts within the secondary classroom activity system across the six-week research cycle.

Quantitative sources included pre- and post-test TTCT scores, SDLI and MAILS Likert-scale scores, and multimodal product rubric scores. Qualitative sources included audio turn-talk records, digital prompt-engineering logs, and semi-structured exit interviews.

Using multiple sources let the study examine both measurable changes in student outcomes and the cognitive and behavioral processes behind human-AI collaboration.

3.3.5 Primary sources

Primary data were collected during active classroom sessions and through direct assessment of participating students and teachers:

1. **Student Pre- and Post-Intervention Assessment Scores:** from the standardized TTCT, SDLI, and MAILS batteries.
2. **Multi-Turn Interaction Logs (Turn-Talks):** recorded histories of the Dutch/English natural language prompts students entered and the corresponding LLM responses, averaging about 16 turns per session.
3. **Creative Multimodal Artifacts:** final student portfolios — digital stories, presentations, product prototypes — evaluated by two independent raters using a standardized rubric.
4. **Semi-Structured Interview Transcripts:** exit interviews with **N = 20** purposively sampled students and all **N = 8** participating teachers.

3.3.6 Secondary sources

Online post published international journal and articles, literatures, newspaper, portals and different organization report. These are the secondary sources of data that provide basic necessary guideline for my research study.

3.4 Data collection procedures

Data collection procedure is a mechanism or way to collect data. I was collected data for my research study in the following way:

a) At first, I have visited the college in Nawalpur District.
b) After that, I have taken permission from administrator and teachers.
c) Next, I have distributed the question papers to teachers who use web 3.0 in the classroom during the teaching learning activates.
d) After that, I have fetched the question paper from them after few days.
e) Finally, I have thanked to teachers, students and administrator for their kind support and help for my research study.

3.5 Data analysis an interpretation procedures

Both open ended and close ended questionnaire ware used in my research. The data analysis methods include both qualitative and quantitative procedures. Qualitative data will measure manually into nominal and ordinal as per Likert scale. Then descriptive analysis of mean, frequency, and percentage of each questions are calculate manually. Tables will create using Microsoft word and Microsoft excel. Then the researcher will insert and analyze collect data descriptively and analytically.

3.6 Ethical considerations

This study will conduct for the academic purpose. While collecting data, ethical considerations will take to ensure for the data privacy. With regarding this study, I will clearly inform my purpose to my respondents. I will not use the data for other purpose except my research work. I will not publish anything that will affect the prestige of respondents. The finding of study will not harm to the respondents. I best possibly 26h prevent physical, psychological and social discomforts of the respondents and building good bond with them. Different scholars' views, idea and definition will proper citation and references according to APA format.

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